



COMP.2800: Software Development

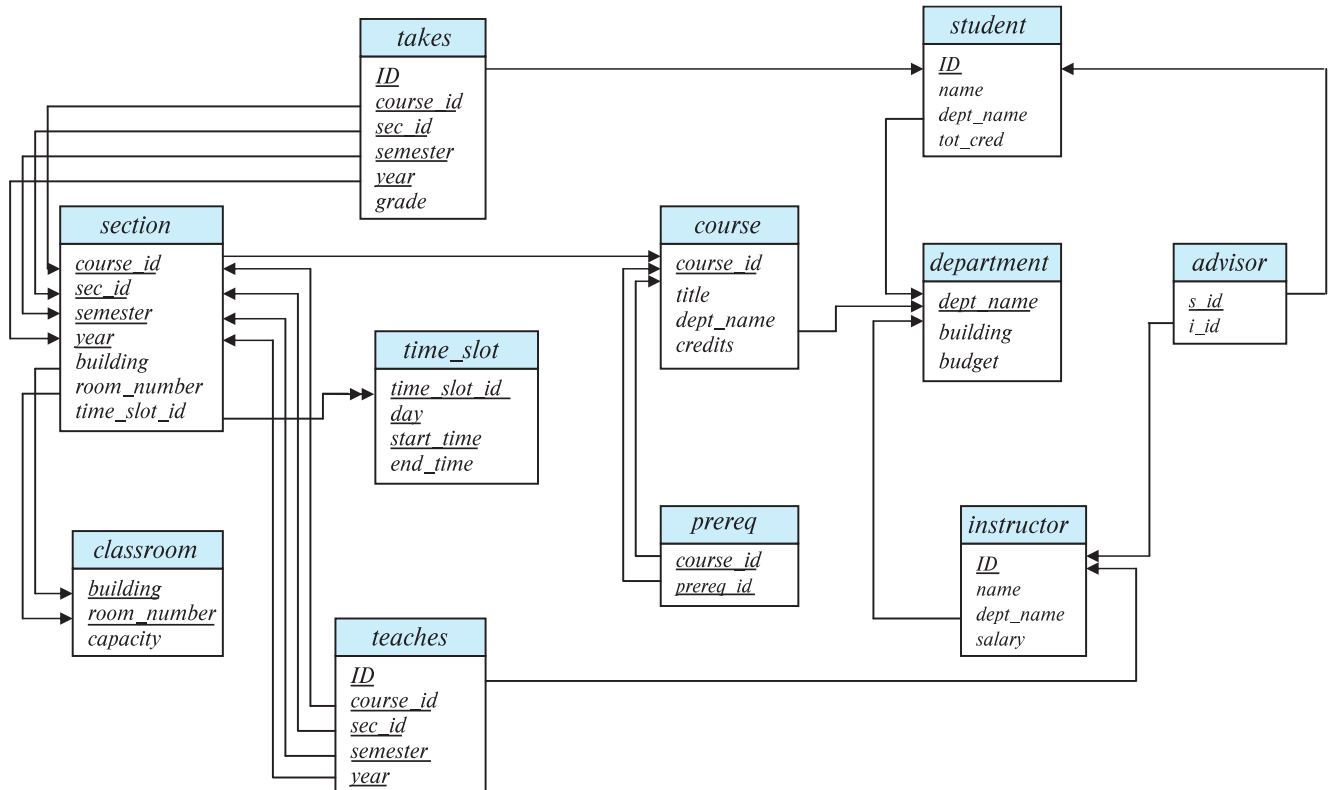
11.1.2. DDL and DML Commands in MySQL

INTRODUCTION

In this tutorial-based lecture you will use the MySQL RDBMS environment you have download, installed, and configured in your laptop to test one the most well-know and widely used databased systems, namely MySQL (<https://www.mysql.com/>).

A. SQL DATA DEFINITION LANNGUAGE (DLL)

1. Use DDL to create the University database shown in the following image (ER Diagram).



2. CREATE a new DATABASE.

```
mysql> CREATE DATABASE my_university;
Query OK, 1 row affected (0.22 sec)

mysql> USE my_university;
Database changed
mysql> SHOW TABLES;
Empty set (0.00 sec)

mysql> 
```



3. CREATE TABLES.

```
mysql> create table classroom
-> (building varchar(15),
-> room_number varchar(7),
-> capacity numeric(4,0),
-> primary key (building, room_number)
-> );
```

Query OK, 0 rows affected (1.28 sec)

```
mysql> create table department
-> (dept_name varchar(20),
-> building varchar(15),
-> budget numeric(12,2) check (budget > 0),
-> primary key (dept_name)
-> );
```

Query OK, 0 rows affected (1.68 sec)

```
mysql> create table course
-> (course_id varchar(8),
-> title varchar(50),
-> dept_name varchar(20),
-> credits numeric(2,0),
-> primary key (course_id),
-> foreign key (dept_name) references department (dept_name)
-> );
```

Query OK, 0 rows affected (1.59 sec)

mysql> █

```
mysql> create table section
-> (course_id varchar(8),
-> sec_id varchar(8),
-> semester varchar(6),
-> year numeric(4,0),
-> building varchar(15),
-> room_number varchar(7),
-> time_slot_id varchar(4),
-> primary key (course_id, sec_id, semester, year),
-> foreign key (course_id) references course (course_id),
-> foreign key (building, room_number) references classroom (building, room_number)
-> );
```

Query OK, 0 rows affected (2.29 sec)

mysql> █

4. Practice yourselves:

- Using the ERD above, create the database and the tables mentioned before.
- Create the following tables according to the ERD:
 - instructor
 - student
 - takes



B. LOAD A MySQL SCHEMA

1. When you have an existing SQL script which defines a schema, you can load it directly into MySQL Server by running this command in your terminal:

```
mysql -u <username> -p new_db < '/path_to_the_sql_file'
```

Example: `mysql -u root -p university_db < 'Desktop/DDL.sql'`

2. The SQL script(s) usually do not include the `CREATE DATABASE` command, so you will have to create an empty database in MySQL before loading the SQL schema.

```
mysql> CREATE DATABASE university_db;  
Query OK, 1 row affected (0.23 sec)
```

```
mysql> █
```

3. All Relations (TABLES) will be loaded into MySQL server.

```
mysql> USE university_db;  
Reading table information for completion of table and column names  
You can turn off this feature to get a quicker startup with -A
```

Database changed

```
mysql> SHOW TABLES;
```

```
+-----+  
| Tables_in_university_db |  
+-----+  
| advisor  
| classroom  
| course  
| department  
| instructor  
| prereq  
| section  
| student  
| takes  
| teaches  
| time_slot  
+-----+
```

```
11 rows in set (0.01 sec)
```

```
mysql> █
```



C. SQL DATA MANIPULATION LANGUAGE (DML)

Use DML commands to insert data into tables in the `university_db` created before.

1. Use this command to insert a single column:

```
INSERT INTO table_name (column1, column2, column3, ...)
VALUES (value1, value2, value3, ...)
```

When inserting values for all the columns in the table and the values added are in the same order in which the columns were created, it is not necessary to specify the column names in the SQL query.

```
mysql> INSERT INTO classroom(building, room_number, capacity) VALUES ('Packard', '101', '500');
Query OK, 1 row affected (0.35 sec)

mysql> INSERT INTO classroom VALUES ('Ryan', '101', '500');
Query OK, 1 row affected (0.16 sec)
```

2. To insert multiple tuples into the database, we will use this `SQLInsertFile` available on the Brightspace, to load some data into the database tables.
3. Download the file and run this command outside `mysql` in the terminal:

```
mysql -u <username> -p db_name <'path_to_the_sql_file'>
```

Example:

```
mysql -u root -p university_db
<'Desktop/smallRelationsInsertFile.sql'>
```
4. Alternatively, you can run this command inside `mysql` client command line:

```
mysql> source path_to_the_sql_file
```
5. To check the data inserted, log into MySQL again and explore some tables.

```
mysql> USE university_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM instructor;
```

ID	name	dept_name	salary
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

```
12 rows in set (0.00 sec)
```




C. SELECT QUERIES

We will now be using the `SELECT` command to query the database(s).

1. Run the following simple `SELECT` queries in MySQL.

```
mysql> SELECT dept_name FROM instructor;
+-----+
| dept_name |
+-----+
| Biology   |
| Comp. Sci. |
| Comp. Sci. |
| Comp. Sci. |
| Elec. Eng. |
| Finance   |
| Finance   |
| History   |
| History   |
| Music     |
| Physics   |
| Physics   |
+-----+
12 rows in set (0.00 sec)
```

```
mysql> SELECT DISTINCT dept_name FROM instructor;
+-----+
| dept_name |
+-----+
| Biology   |
| Comp. Sci. |
| Elec. Eng. |
| Finance   |
| History   |
| Music     |
| Physics   |
+-----+
7 rows in set (0.00 sec)
```

2. Use the keyword `WHERE` to define the criteria of your selection (filter).

```
mysql> SELECT ID, name, salary FROM instructor WHERE dept_name='Comp. Sci.' AND salary > 70000;
+-----+-----+-----+
| ID   | name  | salary |
+-----+-----+-----+
| 45565 | Katz   | 75000.00 |
| 83821 | Brandt | 92000.00 |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

3. Get the monthly salary using the `AS` clause:

```
mysql> SELECT ID, name, salary/12 as monthly_salary FROM instructor WHERE dept_name='Comp. Sci.';
+-----+-----+-----+
| ID   | name      | monthly_salary |
+-----+-----+-----+
| 10101 | Srinivasan | 5416.666667 |
| 45565 | Katz       | 6250.000000 |
| 83821 | Brandt    | 7666.666667 |
+-----+-----+-----+
3 rows in set (0.04 sec)
```

4. Join Query Example: Retrieve the names of all instructors, along with their department names and department building name.



```
mysql> SELECT name, instructor.dept_name, building FROM instructor, department WHERE instructor.dept_name = department.dept_name;
+-----+-----+-----+
| name   | dept_name | building |
+-----+-----+-----+
| Crick   | Biology   | Watson   |
| Srinivasan | Comp. Sci. | Taylor   |
| Katz     | Comp. Sci. | Taylor   |
| Brandt  | Comp. Sci. | Taylor   |
| Kim     | Elec. Eng. | Taylor   |
| Wu      | Finance   | Painter   |
| Singh   | Finance   | Painter   |
| El Said | History   | Painter   |
| Califieri | History   | Painter   |
| Mozart  | Music     | Packard   |
| Einstein | Physics   | Watson    |
| Gold    | Physics   | Watson    |
+-----+-----+-----+
12 rows in set (0.00 sec)
```

5. You can use aliases for table names as follows:

```
SELECT name, A.dept_name, building
FROM instructor A, department B
WHERE A.dept_name = B.dept_name;
```